

STOLEN STRATA

Quantifying the Anthropogenic Erasure of Kashmir's Karewa Terraces and Its Threat to the Saffron Economy

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The Question

The Kashmir Valley's karewas — flat-topped Plio-Pleistocene terraces left behind by an ancient intermontane lake — are being quietly flattened by unregulated soil mining and urban sprawl. The loess capping these terraces is exactly what makes them suitable for saffron (*Crocus sativus*) cultivation, a Geographical Indication-tagged crop central to the Pampore belt's economy. Journalistic accounts document the loss anecdotally; no systematic, multi-decadal, satellite-based quantification had previously connected the physical erasure of this landform to the economic fate of the industry it sustains.

The Method

A fully scripted Python pipeline delineates 201 karewa terrace boundaries from a Copernicus DEM using Topographic Position Index and slope thresholding — no manual digitization. Bare-earth fraction per terrace is tracked across four season-matched time points (1994, 2005, 2015, 2025) via Landsat and Sentinel-2 composites. Saffron-cultivating terraces are identified via an inverted-phenology signature, then tested for proximity to already-degraded terraces. A Mann-Whitney U test checks whether degradation is statistically linked to road accessibility. QGIS is used only for visual quality assurance, never for manual digitization.

The Finding

Degradation is recent and accelerating, not a slow multi-decadal process. Mean bare-earth fraction across all 201 terraces was essentially flat from 1994 to 2015, then more than tripled between 2015 and 2025. Degradation is also spatially concentrated and access-driven: 67% of total loss sits within just 12.4% of terraces, and degraded terraces sit statistically significantly closer to roads than intact ones. Saffron is not yet directly overlapped by degradation, but 43% of detected saffron-cultivating terraces sit within 1 km of an already-degraded terrace — the nearest just 80 m away.

Metric	Value
Terraces delineated (DEM-derived)	201, across 33.05 sq. km
Net bare-earth conversion, 1994–2025	190.3 ha (5.8% of mapped terrace area)
Loss concentration	67% of loss within 12.4% of terraces (25 of 201)
Bare-earth fraction trend	1.84% (1994) → 2.62% (2005) → 2.63% (2015) → 8.43% (2025)
Road-proximity link	75.6 m (degraded) vs. 133.1 m (intact); Mann-Whitney p = 0.0116
Saffron encroachment risk	43% of 14 detected saffron terraces within 1 km of degradation; nearest 80 m

Validation & Robustness Checklist

- ✓ Season-matching (June–Sept, both years) corrected an initial seasonal-mismatch NDVI artefact before it could distort the headline trend.
- ✓ Resolution-artefact tested: resampling 2025 Sentinel-2 (10 m) to match 1994–2015 Landsat (30 m) confirms the acceleration survives — 7.48% vs. a ~2.6% flat baseline.
- ✓ Degradation threshold swept 5–30 points — the flagged terrace count stays stable (23–31) across the 12–20 point neighbourhood.
- ✓ Road-proximity link is statistically significant (Mann-Whitney p = 0.0116), not a visual impression.
- ✓ Saffron proximity-risk finding (43% at 1 km) is stable across a 500–2,500 m threshold sweep and a 0.05–0.175 detection-index sweep — not an artefact of one cutoff.
- ! ~13% of the post-2015 jump's magnitude is attributable to the 10 m→30 m resolution difference — real, quantified, and disclosed rather than folded into the headline number.
- ! Detected saffron area (225.4 ha) sits well below the FAO GIAHS baseline (3,200 ha) — reported openly as a detection-recall limitation of a two-filter pipeline, not as evidence of saffron-area loss.

Honest Limitation

The detected saffron extent is a floor, not a ceiling: compounding two restrictive filters (bare-earth terrace delineation intersected with a saffron spectral-signature threshold) narrows recall well below the independently reported FAO baseline. This is disclosed as a precision-recall trade-off, not presented as a land-loss statistic — and the proximity-risk finding it feeds into is itself shown to be stable across a wide range of detection settings.

Real-World Relevance

The karewas' loess cap is the only reason Kashmir's Geographical Indication-tagged saffron grows here at all — a crop the FAO baseline ties to roughly 17,000 farming families. Every hectare mined for construction aggregate is permanently unavailable to that economy: unlike a fallow field, a flattened terrace cannot be replanted. Because degradation tracks road proximity, this is an actionable finding, not just a descriptive one — targeted permitting and monitoring along road corridors near saffron-adjacent terraces could measurably slow further loss before the post-2015 acceleration reaches deeper into the Geographical Indication belt.